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Project Thesis

submitted in partial fulfillment of the
requirements for the course "Applied Computer Science"

A comparision of various deep learning based detection systems for multi-object tracking

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Abstract

[Put your abstract here.]

General structure of an abstract, write 1 to 2 sentences per section

1. A general statement introducing the broad research area of the particular topic being investigated.
2. An explanation of the specific problem (difficulty, obstacle, challenge) to be solved.
3. A review of existing or standard solutions to this problem and their limitations.
4. An outline of the proposed new solution.
5. A summary of how the solution was evaluated and what the outcomes of the evaluation were.

You may find the following resources useful:

- `https://www.grammarly.com/blog/write-an-abstract/`
- `https://www.editage.com/insights/manuscript-structure-how-to-convey-your-most-im`
- More useful links: `https://hps.vi4io.org/teaching/ressources/start`

Declaration on the use of ChatGPT and comparable tools in the context of examinations

In this work I have used ChatGPT or another AI as follows:

- ☑ Not at all
- ☐ During brainstorming
- ☐ When creating the outline
- ☐ To write individual passages, altogether to the extent of 0% of the entire text
- ☐ For the development of software source texts
- ☐ For optimizing or restructuring software source texts
- ☐ For proofreading or optimizing
- ☐ Further, namely: -

I hereby declare that I have stated all uses completely.
Missing or incorrect information will be considered as an attempt to cheat.

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HPC High-Performance Computing

1 Introduction

2 Related Work

2.1 Overview of modern detection systems

Edozie et. al [1] provides a review across the past developments of object detection techniques in the past 26 years. Besides describing the evolution of deep learning based object detection systems they also compare them to the classical non-deep-learning based object detection methods and algorithms.

They provide a wide overview of datasets, methods and research that was important in the past 26 years. Beside providing classical object detection methods they also present an overview of modern deep learning based object detector which should be summarized here to give an impression of the state of the art.

Object detection systems are separated in One-Stage and Two-Stage object detectors. Where the two-stage detector first generates proposals of the detection and then encodes them to make class predictions. Otherwise the first-stage detector doesn't consider region proposals, they simply rely on the locations with the most objects and either classify them as an object or background. Through this simplification One-Stage detectors are generally faster than two-stage detectors. In table **??** you will find a list of modern two-stage detectors and their implementend advancment based on their disadvantage as presented by [1].

Table 1: Two-stage detection systems and their advancemnets

Year	**Method**	**Advancement**
2011	R-CNN	Selective search for region proposal extracted and input to a Convolutional Neural Network.
2015	SPPNet	Applies Spatial Pyramid Pooling to have many pooling layers for different scales
2015	Fast R-CNN	Use Region of Interest pooling layer to share region proposals of convolutions
2016	Faster R-CNN	Incooperate a Regional Proposal Network eliminating the need of seperate region proposal algorithms.
2017	FPN	Combines shallow and deep layer features to enable detection of objects at various scales.
2017	Mask R-CNN	Adding a mask network branch for RoI segment prediction, enabling simultaneous object detection and instance segmentation
2018	PANet	Adds an extra upward PANet above the FPN.
2019	TridenTNet	employs a quick inference technique with just one main branch.
2020	SpineNet	cale-permuted intermediate characteristics and cross-scale connections.

Neha et al. [2] provides an overview of detection systems using deep learning and classical approaches. They provide a summary of datasets, metrics and methods relevant in this domain. Further they provide a list of strengths and weaknesses of the different

deep-learning based detection systems and also provide benchmark results on various detection systems, e.g.: R-CNN, Yolo etc.

2.2 Integration of detection systems in tracking systems

3 Benchmark comparision of various detection systems

3.1 Benchmark Environment

3.2 Object detector implementations

3.3 Detector-Tracker integrations

3.4 Evaluation

4 Results and Analysis

5 Discussion

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References

[1] E. Edozie, A. N. Shuaibu, U. K. John, *et al.*, "Comprehensive review of recent developments in visual object detection based on deep learning," *Artif Intell Rev*, vol. 58, p. 277, Jun. 2025. DOI: 10.1007/s10462-025-11284-w. [Online]. Available: https://doi.org/10.1007/s10462-025-11284-w.

[2] F. Neha, D. Bhati, D. K. Shukla, and M. Amiruzzaman, *From classical techniques to convolution-based models: A review of object detection algorithms*, 2024. arXiv: 2412.05252 [cs.CV]. [Online]. Available: https://arxiv.org/abs/2412.05252.

A Additional Materials